AI Exposure and Wage Growth in Australia

2020–2024

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ECC3479 Empirical Project
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Research question

How is occupational AI exposure associated with wage growth in Australia over 2020–2024, and how has that association changed year by year?

Why it matters

- AI can *substitute* for tasks in high-exposure occupations → lower labour demand, compressed wages
- AI can *complement* tasks → higher productivity, potentially higher wages (Autor, 2015)
- Which channel dominates is an open empirical question
- Most existing evidence comes from the US — a direct Australian estimate is independently valuable

Data

Individual-level: HILDA Survey

- Melbourne Institute, annual since 2001
- Nationally representative longitudinal household survey
- Key variables: annual wages, 2-digit ANZSCO occupation code, education, age, sex, state

AI exposure: Felten et al. (2021) AIOE

- Continuous score per US occupation (SOC-2010)
- Rates relevance of 52 AI application areas to O*NET task requirements
- Mapped to Australia via SOC $\rightarrow$ ISCO-08 $\rightarrow$ ANZSCO 2-digit crosswalk
- Covers 99.6% of valid occupation–year observations

Sample construction

Step	Persons
All HILDA respondents, 2020–2024	14,399
Drop missing wage / AI score / education	14,296
Drop without valid 2020 observation	8,864
Drop fewer than two years observed	7,913
Drop unmappable education code	7,912
Final estimation sample	**7,912**

Final panel: 7,912 individuals $\times$ up to 5 years $=$ **32,664 person-year observations**

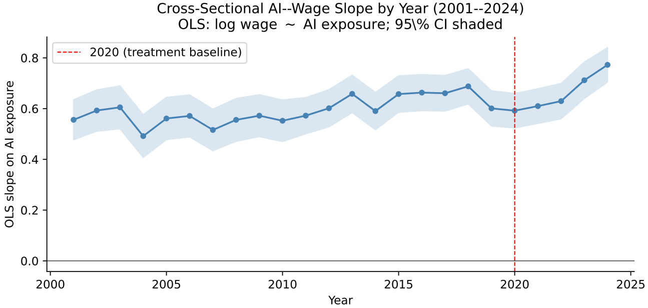
AI exposure is fixed to each worker's **2020 occupation** and held constant throughout.

Two-Way Fixed Effects (TWFE) event-study model:

$$\log w_{it} = \sum_{t \in \{2021, 2022, 2023, 2024\}} \beta_t \cdot \left(\mathrm{AI}_i \times \mathbf{1}\{\text{year} = t\}\right) \; + \; \gamma \, \mathrm{edu}_{it} \; + \; \mu_i \; + \; \lambda_t \; + \; \varepsilon_{it}$$

- AI_i = 2020 baseline AIOE score (continuous; held fixed)
- μ_i = individual fixed effects $\quad \lambda_t$ = year fixed effects
- Standard errors clustered at individual level
- $\hat{\beta}_t$ = conditional association between AI exposure and within-person wage growth in year t relative to 2020

Identification assumption and design choice



Source: output/robustness_checks.ipynb

Parallel trends: absent a differential AI shock, high- and low-exposure workers would have followed similar wage trajectories.

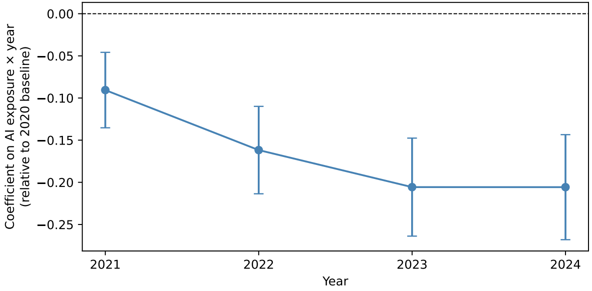
Evidence: cross-sectional OLS slope of wages on AI exposure, year by year 2001–2019. The slope is stable at ≈ 0.59 throughout — no pre-existing trend.

Why fix exposure to 2020?

- 37.5% of workers change occupation during 2020–2024
- Updating annually conflates wage effects with job-changers' selection into higher-paying roles
- Fixing to 2020 removes this endogenous sorting bias

Main results

Figure 1: TWFE Event-Study — AI Exposure and Wage Growth (2020–2024)
95% confidence intervals; clustered SE at individual level



Source: output/primary_analysis.ipynb
The gap **grew** monotonically 2021 $\rightarrow$ 2023, then **held steady** in 2024.

Finding: Workers in higher-AI occupations fell behind in wage growth every year from 2021 to 2024.

Year	$\hat{\beta}$	wage growth gap
2021	-0.091	-2.1%
2022	-0.162	-3.8%
2023	-0.206	-4.8%
2024	-0.206	-4.8%

All $p < 0.001$ (clustered SE)
gap $\approx \hat{\beta} \times 0.234$

Economic significance

- Average annual wage growth in the sample: $\approx \mathbf{8.5\%}$/year
- The **4.8 percentage-point gap** is roughly $\mathbf{0.6}\times$ that average — high-AI workers fall behind by **more than half** a typical annual pay rise, conditional on their own prior trajectory and the economy-wide wage trend
- The gap widened **monotonically** from 2021 (-0.091) to 2023 (-0.206), then **held steady** in 2024 (-0.206) — a persistent divergence, not a one-off adjustment

Robustness

Passed (2022–2024 negative and significant):

- HC3 heteroskedasticity-robust SE
- Dropping education control
- Adding state-of-residence fixed effects
- Wage trimming (top/bottom 1%)
- IHS functional form
- Alternative base year (2021)
- Age-as-outcome placebo ≈ 0

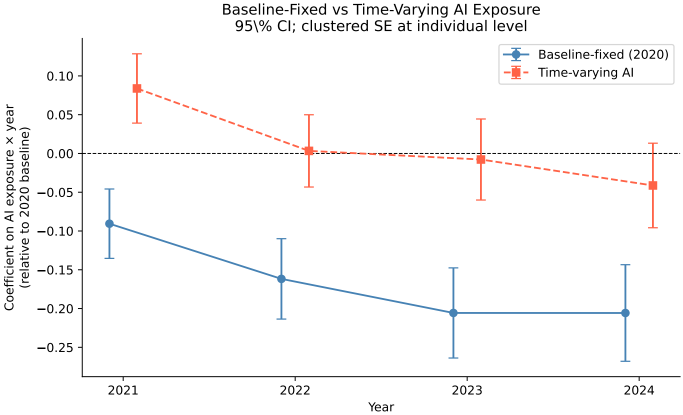
Fragility:

- **Time-varying AI scores** (occupation switching)

  Including switching shifts all coefficients upward, but the 2022–2024 downward trend holds in both specifications
- **Restricting to non-switchers**

  Effect shrinks but stays negative — the result is not entirely driven by occupation switching

Time-varying AI scores (occupation switching)



Source: output/robustness_checks.ipynb. Circles: main specification (AI fixed to 2020). Squares: time-varying (AI updates annually). The 2021 sign flip under time-varying exposure reflects endogenous job-switching. 2022–2024 negative trend holds in both specifications.

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Limitations

Most important threat:

- Pandemic demand temporarily inflated wages in AI-adjacent occupations in exactly the year used as the baseline
- The post-2020 compression may partly reflect that premium reversing rather than AI-driven substitution
- Alternative base year (2021) confirms compression persisted beyond the pandemic adjustment — but the design **cannot decompose** the two mechanisms

Other threats:

- HILDA covers formal employment only — casual and freelance workers are excluded

Conclusion and next steps

What we can conclude:

- A descriptive finding: higher AI exposure is negatively associated with wage growth in Australia over 2020–2024
- The gap reaches ≈4.8 percentage points by 2023 and is persistent through 2024
- The pattern is robust across all specification checks and holds for occupation non-switchers

What we cannot conclude:

- Whether this is AI-driven substitution, a pandemic premium reversal, or both

What would resolve it:

- Use ChatGPT's rollout (late 2022) as a natural experiment — separates AI adoption from the pandemic
- Extend HILDA to 2027 — the further from 2020, the less the pandemic baseline matters

Full report and replication code:
https://github.com/TingyingHuang/ECC3479-The-impact-of-AI-exposure-on-wage-growth

References

- Autor, D. H. (2015). Why are there still so many jobs? The history and future of workplace automation. *Journal of Economic Perspectives*, *29*(3), 3–30. https://doi.org/10.1257/jep.29.3.3
- Felten, E., Raj, M., & Seamans, R. (2021). Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, *42*(12), 2195–2217. https://doi.org/10.1002/smj.3286
- Acemoglu, D., & Restrepo, P. (2018). *Artificial intelligence, automation, and work* (NBER Working Paper No. 24196). National Bureau of Economic Research. https://doi.org/10.3386/w24196
- Angrist, J. D., & Pischke, J.-S. (2009). *Mostly harmless econometrics: An empiricist's companion*. Princeton University Press.
- Melbourne Institute. (2024). *Household, Income and Labour Dynamics in Australia (HILDA) Survey* [Dataset]. University of Melbourne.